Tutorial 2 — Complete Solutions

Multiple Regression and Partial Effects

Introductory Econometrics — Télécom Paris

Based on the supplied Tutorial 2 handout and Python lab

Scope. Answers follow the order and notation of the supplied `exercises.pdf`. Numerical results were obtained by running the completed `code/lab2.py` on the original `data/WAGE1.raw` (526 workers). The script reproduces the tables and figures in `code/output/`.

1 Theory Exercise 1 — What does a higher R^2 establish?

On the same n observations, consider the nested models

$$\text{A: } y_i = a_0 + a_1 x_{1i} + u_i, \qquad \text{B: } y_i = b_0 + b_1 x_{1i} + b_2 x_{2i} + v_i.$$

Both include an intercept and have full column rank. Denote the residual sums of squares by RSS_A and RSS_B.

(a) Show that $RSS_B \leq RSS_A$

The parameter vector $(a_0, a_1, 0)$ is feasible in Model B and reproduces every fitted value from Model A. Since OLS minimizes the residual sum of squares over all coefficients of Model B,

$$RSS_B = \min_{b_0,b_1,b_2} \sum_i (y_i - b_0 - b_1 x_{1i} - b_2 x_{2i})^2 \leq \sum_i (y_i - \widehat{a}_0 - \widehat{a}_1 x_{1i})^2 = RSS_A.$$

$\boxed{RSS_B \leq RSS_A}$: Adding a regressor cannot worsen the in-sample OLS fit when the sample stays the same.

(b) Compare the two R^2 values; equality condition

With an intercept and a common outcome/sample, $TSS = \sum_i (y_i - \bar{y})^2$ is identical for both models. Therefore,

$$R_B^2 = 1 - \frac{RSS_B}{TSS} \geq 1 - \frac{RSS_A}{TSS} = R_A^2.$$

Equality holds if and only if $RSS_B = RSS_A$. Under the stated full-rank assumption, this occurs exactly when the estimated coefficient of the added regressor is zero: $\widehat{b}_2 = 0$ (equivalently, the added variable has no sample partial contribution after controlling for the intercept and x_1).

$\boxed{R_B^2 \geq R_A^2, \qquad R_B^2 = R_A^2 \iff \widehat{b}_2 = 0.}$

(c) Why a larger R^2 does not imply a better causal model

R^2 measures the fraction of sample outcome variation explained by the fitted linear model. It does *not* test whether the regressors are exogenous, whether confounders are omitted, or whether a control is a mediator/collider. Adding an irrelevant or endogenous variable can raise R^2 without making a causal interpretation valid. Causality requires substantive identification assumptions, especially an appropriate zero-conditional-mean assumption.

(d) Why adjusted R^2 can fall; derive the condition

For k slope regressors plus an intercept,

$$\overline{R}^2 = 1 - \frac{RSS/(n-k-1)}{TSS/(n-1)}.$$

Model A has $k_A = 1$ and Model B has $k_B = 2$. Thus,

$$\begin{aligned}\overline{R}_B^2 \geq \overline{R}_A^2 &\iff \frac{RSS_B}{n-3} \leq \frac{RSS_A}{n-2} \\ &\iff \boxed{\frac{RSS_B}{RSS_A} \leq \frac{n-3}{n-2}} \\ &\iff \boxed{RSS_A - RSS_B \geq \frac{RSS_A}{n-2}}.\end{aligned}$$

For $RSS_B > 0$, the equivalent partial F statistic for adding one regressor is

$$F = \frac{(RSS_A - RSS_B)/1}{RSS_B/(n-3)} \geq 1.$$

Adjusted R^2 falls if the reduction in RSS is too small to offset the loss of one residual degree of freedom. This is an *in-sample penalized-fit* criterion, not a causal-validity test.

2 Python Lab 1 — Does adding controls solve the education–wage problem?

Sample and variables: $n = 526$ workers from the historical US cross-section; *wage* is hourly earnings in 1976 US dollars; *educ*, *exper*, *tenure* are in years. Both regressions use the same 526 observations.

$$\text{Model A: } \log(wage_i) = a_0 + a_1 educ_i + v_i,$$
$$\text{Model B: } \log(wage_i) = b_0 + b_1 educ_i + b_2 exper_i + b_3 tenure_i + u_i.$$

2.1 Step 1 — Define the comparison before estimating

(a) Which workers are compared? What is fixed? Is it causal?

Model B compares workers who have the same experience and tenure but different education levels. The coefficient b_1 is a *partial association* holding *exper* and *tenure* fixed. This comparison is not automatically causal: unobserved ability, family background and other factors may jointly influence education and wages.

(b) Predict the omitted-variable direction.

The sample correlations are

$$\text{Corr}(educ, exper) = -0.299542, \quad \text{Corr}(educ, tenure) = -0.056173, \quad \text{Corr}(exper, tenure) = 0.499291.$$

If both experience and tenure have positive *partial* wage effects, and each has a negative slope in its auxiliary regression on education, omitting them would tend to bias the simple education slope *downward*. The correlation signs are suggestive, not conclusive by themselves, especially with multiple omitted regressors. The precise bias depends on the relevant auxiliary slopes and conditional effects.

2.2 Step 2 — Estimate and explain the coefficient change

The completed Python script estimates both models with `statsmodels.formula.api.ols`.

Quantity	Model A	Model B
Education coefficient	0.082744	0.092029
Experience coefficient	—	0.004121
Tenure coefficient	—	0.022067
Observations	526	526
R^2	0.185806	0.316013

(a) How much does the education coefficient change? Interpretation.

$$\Delta\widehat{b}_{educ} = 0.092029 - 0.082744 = \boxed{+0.009285}.$$

That is an increase of about 0.929 percentage points in the approximate semi-log interpretation. In Model B, one additional year of education is associated with **approximately 9.20% higher hourly wages**, holding experience and tenure fixed. The exact fitted proportional comparison for one year is

$$100(e^{0.092029} - 1) \approx \boxed{9.64\%}.$$

These are associations, not identified causal returns.

(b) Does the direction match the prediction?

Yes. The education slope increases from Model A to B, consistent with the prediction that omitting positively wage-associated experience and tenure can push the simple slope downward. Model A's error contains their wage contributions; because the controls and education are correlated, the simple education slope also absorbs part of their association with wages.

(c) Why must $R^2_B \geq R^2_A$? Does this establish exogeneity?

Model A is nested in B and both use the same sample. OLS can always choose zero coefficients for the extra controls, so $RSS_B \leq RSS_A$ and $R^2_B \geq R^2_A$. The observed rise from 0.186 to 0.316 does *not* establish $\mathbb{E}(u \mid educ, exper, tenure) = 0$.

2.3 Step 3 — Audit exogeneity as part of the analysis

(a) What does zero conditional mean require? Three plausible confounders.

$$\boxed{\mathbb{E}(u \mid educ, exper, tenure) = 0.}$$

At every combination of education, experience and tenure, the mean of unobserved wage determinants must be zero. Potential confounders include (i) *ability*: can affect schooling success and productivity; (ii) *parental resources/background*: can affect access to education and job opportunities; and (iii) *local labor-market opportunities*: may affect schooling decisions and wages. Unmeasured ability is particularly concerning because it is difficult to capture fully with observed controls.

(b) Parents' education, a pre-school test score, and current occupation.

Parents' education and a test score measured *before* schooling may proxy for pre-existing family resources or ability. Current occupation may instead be an outcome of education and a channel through which education affects wages. Controlling for occupation can therefore remove part of a total education effect.

(c) Why not add every variable? Give two reasons.

1. **Mediators and timing:** occupation, and possibly some measured experience or tenure, may be affected by education. Conditioning on them can change the estimand from a total to a direct effect.

2. **Selection/collider bias:** controlling for a variable influenced by both schooling and unobserved wage determinants can create spurious conditional associations. Controls must be selected using a causal rationale, not only predictive fit.

(d) Three-sentence conclusion.

Holding experience and tenure fixed, one additional year of education is associated with approximately 9.20% higher hourly wages. The coefficient increases relative to the education-only model because omitted experience and tenure are associated with education and wages. Unobserved ability and family background remain major threats to a causal interpretation.

3 Python Lab 2 — What partialling out does and does not show

Use experience and tenure as the controls. Define $Z = [\mathbf{1}, exper, tenure]$ and $x = educ$, $y = \log(wage)$.

(a) Regress education on the controls and save r_x

Estimate $educ_i = \gamma_0 + \gamma_1 exper_i + \gamma_2 tenure_i + r_{xi}$. A **positive education residual** means a worker has more education than predicted by their experience and tenure under this fitted linear relationship.

(b) Apply Lecture 2, slide 14

By the Frisch–Waugh–Lovell result,

$$\boxed{\widehat{b}_1 = \frac{\sum_i r_{xi} \log(wage_i)}{\sum_i r_{xi}^2}}.$$

The code computes this using the `@` operator for inner products:

```
lecture_slope = float(
    education_residual @ df["log_wage"]
    / (education_residual @ education_residual)
)
```

Numerically,

$$\underbrace{0.092028987319613}_{\text{slide-14 formula}} \approx \underbrace{0.092028987319615}_{\text{Model B OLS coefficient}}.$$

Only the part of education *not linearly predicted* by experience and tenure identifies the education slope in Model B.

(c) Residualize log wage as well; regress r_y on r_x

Regress y on Z and save r_y. Then regress r_y on r_x with an intercept. Because OLS residuals from models with intercepts have mean zero, the residual-on-residual intercept is zero up to numerical precision:

$$\widehat{\alpha} = -7.54 \times 10^{-16} \approx 0, \qquad \widehat{\delta} = 0.092028987319614.$$

The slope is the same as in (b) and Model B because r_x is orthogonal to Z, so $\sum_i r_{xi} y_i = \sum_i r_{xi} r_{yi}$. This is an **algebraic OLS identity**; it does *not* provide evidence for $\mathbb{E}(u \mid educ, exper, tenure) = 0$.

4 Python Lab 3 — When a precise regression answers the wrong question

The professor's Monte Carlo script simulates $n = 150$ observations and 3,000 repetitions for each correlation $\rho \in \{0.2, 0.99\}$ from

$$y = 1 + x_1 + x_2 + u, \quad \mathbb{E}(u \mid x_1, x_2) = 0, \quad \mathrm{Corr}(x_1, x_2) = \rho.$$

The *full* regression includes both x_1 and x_2; the *short* regression omits x_2.

(a) Predict the centres before running

The true full-model partial effect of x_1 is $\beta_1 = 1$. The omitted-variable formula for the short regression gives

$$\operatorname{plim} \widehat{\beta}_{1,short} = \beta_1 + \beta_2\delta = 1 + 1 \cdot \rho = \boxed{1+\rho},$$

where $\delta = \rho$ is the population slope from regressing x_2 on x_1 (both have unit variance). The predicted short-model centres are 1.2 for $\rho = 0.2$ and 1.99 for $\rho = 0.99$.

(b) Compare full-model mean and standard deviation

Simulation statistic	$\rho = 0.20$	$\rho = 0.99$
Mean of full $\widehat{\beta}_1$	0.998334	1.003532
SD of full $\widehat{\beta}_1$	0.084088	0.594605
Mean of full $\widehat{\beta}_2$	1.001549	0.995563
SD of full $\widehat{\beta}_2$	0.084718	0.595313
Corr(full $\widehat{\beta}_1$, full $\widehat{\beta}_2$)	-0.217298	-0.990120
SD of $\widehat{\beta}_1 + \widehat{\beta}_2$	0.105603	0.083638
Mean of short slope	1.198236	1.989213
SD of short slope	0.112801	0.084060

The full-model coefficient stays centred close to 1 in both designs. Its sampling SD increases sharply with ρ because x_1 has little residual variation after controlling for x_2. **High multicollinearity increases variance, not bias**, under the correctly specified exogenous full model.

(c) Why are the two full-model slopes negatively related? Why is their sum stable?

When $\rho \approx 1$, x_1 and x_2 move almost together. A larger estimated coefficient on x_1 can be offset by a smaller one on x_2 with little change in fitted y. This creates a strong negative correlation (-0.9901) between coefficient estimates. Their *sum* is more stable because the combined effect along the shared direction is well identified, even though the two separate effects are imprecise.

(d) Compare full and short models at $\rho = 0.99$

The short slope is much more precise (SD 0.0841) than the full slope (SD 0.5946), but it is centred near 1.99, not the true partial effect 1. It estimates the combined association of x_1 with y, including the omitted x_2 contribution. **Precision around the wrong target is not accuracy for the partial effect.**

(e) Should x_2 be deleted to reduce the standard error?

No, not if the goal is to estimate the partial effect of x_1 and x_2 truly belongs in the model. Removing x_2 reduces variance but introduces omitted-variable bias. The choice of controls should follow the estimand and identification assumptions; a small standard error is not sufficient justification.

Simulation figure

The completed Python script saves `code/output/lab3_multicollinearity.png`. If the figure exists in the project, it is included below.

Run code/lab2.py to generate the simulation figure.

5 Run instructions

Keep the supplied structure:

```
tutorial-2/
  exercises.pdf
  data/WAGE1.raw
  code/lab2.py
  code/output/                    (generated automatically)
  report/tutorial2_complete.tex
```

Install the Python requirements with `python -m pip install -r requirements.txt` from the `tutorial-2` directory. Run `python code/lab2.py`. Compile this report from the `report/` directory using `pdflatex tutorial2_complete.tex` twice, or upload the `.tex` file to Overleaf with the generated image if desired.